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A DSA Algorithm for Mortality Forecasting

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Borrowing information from populations with similar structural mortality patterns and trajectories has been well recognized as an useful strategy to the mortality forecasting of a target population. This article presents a flexible framework for the selection of populations from a given candidate pool to assist a target population in mortality forecasting. The defining feature of the framework is the deletion-substitution-addition (DSA) algorithm, which is entirely data driven and versatile to work with any multiple-population model for mortality prediction. In numerical studies, the framework with an extended augmented common factor model is applied to the Human Mortality Database, and the superiority of the proposed framework is evident in mortality forecasting performance.

1. INTRODUCTION

A comprehensive understanding and quantification of human mortality rates is the fundamental task for actuaries because a reliable mortality forecast is crucial for the pricing and valuation of various life insurance and living benefits products. An inaccurate forecast can expose providers to an imbalance between assets and liabilities. In the case of life insurance, the benefits of which are typically payable upon the occurrence of death, an underestimate of the future mortality rates can cause the insurance provider to suffer the consequences of underpricing. In the cases of living benefits such as annuities and pension plans, an underestimate of future mortality rates can lead to overpricing, thereby causing the provider to lose a competitive edge on the market. Further, an overestimate can lead to an inadequate premium charge, and the providers may eventually have to increase reserves to reflect the actual mortality rates, thus creating potential insolvency risk.

Mortality prediction presents challenges to actuarial research as the population experiences unprecedented longevity. Numerous models for mortality modeling and forecasting have been developed. The Lee-Carter mortality projection model, introduced by Lee and Carter (1992), has been regarded as the prototype of extrapolative mortality forecasting methods. In their model, the logarithmic age-specific death rates (ASDRs) for a single population over a specified period are decomposed into the age pattern of mortality, the overall time trend, and the age-specific differences in the context of the time trend.

Actuarial researchers and demographers have developed various extensions of mortality forecasting models since Lee and Carter's (1992) model. For example, variants of the Lee-Carter model with alternative estimation procedures or adjustments on assumptions have been proposed to solve the deficiencies of the Lee-Carter model. Wilmoth (1993) introduced the weighted least squares estimation in the presence of zero mortality rates and nonconstant variance in mortality data. Lee and Miller (2001) and Booth, Maindonald, and Smith (2002) took issue with the jump-off bias and put forward different approaches to adjust the overall time trend. de Jong and Tickle (2006) generalized the model under the state-space framework to encompass a wide range of flexible multivariate time series models, among which the Lee-Carter model is a special case. Renshaw and Haberman (2006) extended the Lee-Carter model through a cohort parameter to describe the nonlinear mortality trend in order to improve its forecasting performance. To better capture the mortality trend in older ages, Cairns, Blake, and Dowd (2006) introduced the Cairns-Blake-Dowd model, which employs linear or quadratic functions of age to model the logit of the death probabilities at older ages.

In addition to the adjustments made to the Lee-Carter model, new methodologies for mortality modeling have also been proposed in the literature. Currie, Durban, and Eilers (2004) suggested using P-splines for the smoothing and forecasting of two-dimensional mortality tables. Hyndman and Shahid Ullah (2007) estimated mortality rates by treating their realizations as

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functional data. Brouhns, Denuit, and Vermunt (2002) substituted the singular value decomposition (SVD) approach with a log-bilinear Poisson regression procedure for the calibration of the Lee-Carter model. Renshaw and Haberman (2003) reinterpreted the Lee-Carter model and introduced a methodology based on generalized linear modeling. Hatzopoulos and Haberman (2009) used a Poisson regression for the actual number of deaths to build a dynamic parametric model that adopts a logarithmic link function and an orthonormal polynomial predictor structure in age effects. Hatzopoulos and Haberman (2011) incorporated the cohort effect into the model of Hatzopoulos and Haberman (2009) for the analysis of mortality structures.

All of the models and methodologies mentioned above are intended for the forecasting of mortality rates of a single population. When it comes to the forecast of multiple populations, these models treat each population separately and provide a forecast for each population independently. In recognition of the common features in mortality development patterns across populations, coherent multipopulation mortality forecasting methods have been developed and their capacity to enhance forecasting accuracy has been well attested. The prototype of such methods is the augmented common factor (ACF) model proposed by Li and Lee (2005), which jointly models the logarithmic ASDRs of a group of different populations with a common factor and, at the same time, it includes population-specific components to allow disparate development of mortality for individual populations.

Since the introduction of the ACF model, many other multiple-population mortality models have been proposed. In particular, Cairns et al. (2011) and Dowd et al. (2011) discussed the incorporation of the age–period–cohort structure into a two-population modeling framework. Kleinow (2015) proposed the common age effect model in which age has the same effect on the centralized logarithmic mortality rates for all populations in the joint model. Hyndman, Booth, and Yasmeen (2013) and Shang and Hyndman (2017) proposed a functional data-based approach. Russolillo, Giordano, and Haberman (2011) incorporated a third component that represents differences in populations into the decomposition of the logarithmic mortality rates, thereby extending the Lee-Carter model to a three-way structure. Hatzopoulos and Haberman (2013) employed a common sparse age–period model structure for mortality experience under a generalized linear model framework and utilized multiple principal components to extract mortality dynamics. Villegas et al. (2017) offered a comprehensive review of multiple-population models and a systematic assessment that appraises the suitability of available two-population mortality models for the assessment of basis risks resulting from using longevity swaps for pension scheme de-risking.

One crucial step before the application of any multiple-population model for mortality forecasting is the selection of a "proper" group of populations because the information underlying the mortality dynamics of the selected populations can help to enhance the mortality prediction accuracy for a target population. Most of the existing joint mortality modeling procedures assume that the grouping information is either known a priori or prespecified according to certain exogenous information, such as geographic proximity, or socioeconomic variables. However, such a heuristic specification fails to guarantee that every selected population is indeed conducive to an enhancement in mortality prediction for the target population. Cluster analysis, insofar as we can tell, is the only structured method adopted in the literature for the purpose. Hatzopoulos and Haberman (2013) adopted fuzzy *C*-means cluster analysis to select populations with similar mortality characteristics for joint modeling. Fuzzy *C*-means yields a list of cluster centers and a matrix to indicate the level of association each data element is with each center.

In this article, we propose a novel flexible and effective method for the selection of populations. Like cluster analysis, instead of relying on geographic, socioeconomic, or any other exogenous information for grouping, our method is built on a well-designed deletion–substitution–addition (DSA) algorithm, which is entirely data driven and directly "learns" for the best group of populations from a given candidate pool without resorting to any explanatory data input. The DSA algorithm simultaneously derives joint modeling results for the target population and the selected group of populations. We then extrapolate the established model into future periods to obtain mortality forecast for the target population.

The implementation of our proposed prediction framework entails the specification of a multiple-population mortality model and the application of the designed DSA algorithm. Our framework is flexible in that it can be applied with any multiple-population joint models, although an extended ACF model of Li and Lee (2005) is applied for illustrative purposes. As its name implies, our DSA algorithm involves the three moves, deletion, substitution, and addition; each step of the dynamic procedure guides us toward the determination of the optimal group. The algorithm starts from an individual prediction model (derived as a marginal model of the adopted multiple-population model and applied to the target population for prediction) and then applies the three moves in each step of the procedure to develop a sequence of groups in increasing size. A validation step is then consequently applied to select the "best" group among the obtained sequence.

Our DSA algorithm is inspired by the partDSA (see Molinaro, Lostritto, and van der Laan 2010), a DSA algorithm originally designed for prediction by recursively partitioning the covariate space in a regression context. The theoretical foundations of the partDSA are described in Sinisi and van der Laan (2004). Our DSA algorithm is essentially different from the partDSA although both algorithms involve the same three moves in the implementation procedures. The partDSA applies the three

moves to the covariate variables of a regression problem in order to form the optimal partition of the covariate space that is used to establish an efficient nonparametric prediction rule. In contrast, the three moves in our DSA algorithm are applied to the candidate pool of populations for the selection of the optimal group for joint modeling with the target population.

As mentioned earlier, a fuzzy clustering method has been applied by Hatzopoulos and Haberman (2013) to the selection of populations for joint mortality modeling. As a standard data mining tool, cluster analysis has also been applied to various other actuarial topics (e.g., Campbell 1986; Gan and Valdez 2016, 2020; Yao 2016; O'Hagan and Ferrari 2017). As a typical clustering method, our DSA algorithm can also associate a target population with a particular group of populations for joint modeling, but with a substantial difference. At the outset of the implementation, one needs to designate a target population for the DSA algorithm to work toward the selection of populations for joint modeling. The best mortality prediction accuracy the DSA algorithm aims to achieve is exclusively for the target population. If the target population changes to another in the pool, a rerun of the DSA algorithm is necessary. In general, the resulting grouping does not remain unchanged. In contrast, a typical clustering algorithm does not have a target population, and it aims at obtaining a grouping of objects that are more homogeneous within each group (also called cluster) than those across groups.

We apply our DSA based prediction model to the Human Mortality Database (HMD) for 60 gender-specific populations from 30 countries or regions in the world. We compare the performance of the DSA-based model with that of the Lee-Carter model and the ACF models that make use of different grouping strategies. A simple all-in-one strategy, a grouping strategy based on geographic proximity information, as well as grouping obtained by the *k*-means and the *k*-median cluster methods are respectively adopted in the implementation of the ACF models. Using data from 1970 to 2002, we train the models and extrapolate them to the period of 2003–2010 for prediction. The performance is measured by four metrics; that is, the mean squared error, mean absolute error, mean squared percentage error, and mean absolute percentage error. Compared with the benchmark models mentioned earlier, our DSA-based model proves to have provided results with higher prediction accuracy. Summary statistics as well as a formal hypothesis test, the Diebold-Mariano test, on the prediction results for the 60 populations corroborate the superiority of our DSA-based model over the benchmarks.

The rest of this article is organized as follows. Section 2 provides a review of the Lee-Carter and the ACF models and also introduces an extended ACF model. Section 3 gives a detailed account of the DSA algorithm and the DSA-based framework for mortality forecasting. Section 4 presents the numerical studies. Section 5 offers some further discussion and concluding remarks. The Appendix describes the geographic grouping information used for one benchmark ACF model.

2. MORTALITY FORECASTING MODELS

In this section, we first review the Lee-Carter model proposed by Lee and Carter (1992) and the ACF model by Li and Lee (2005) and then propose an extended ACF model to be used with our DSA algorithm.

2.1. Lee-Carter Model

Let $m_{x,t}$ denote the central death rate, also known as the age-specific death rate, for age $x \in \{0, 1, ..., \omega\}$ and year $t \in \{0, 1, ..., T\}$ of a population, where ω represents the limit age of the population. Lee and Carter (1992) summarized the age–period surface of logarithmic ASDRs in the following form:

$$\log m_{x,t} = a_x + b_x k_t + \epsilon_{x,t}, \tag{1}$$

where b_x captures the relative speed of change in mortality at each age x, k_t is an index of the level of mortality for year t, and $\epsilon_{x,t}$ represents the white noise term. The parameterization in Equation (1) is not unique. To avoid the unidentifiability issue with the parameters, normalization constraints $\sum_x b_x = 1$ and $\sum_t k_t = 0$ are imposed, and under these constraints, a_x becomes the average age profile of mortality; that is, the average of logarithmic ASDRs over the modeling time period:

$$a_x = \frac{1}{T+1} \sum_{t=0}^{T} \log m_{x,t}. \tag{2}$$

The above Lee-Carter model is typically calibrated by a two-stage procedure. In the first step, a_x is obtained through Equation (2), b_x and k_t are consequently derived via an SVD procedure, which is essentially a method to use the product of two vectors to approximate the matrix with entries $(\log m_{x,t} - a_x), x \in \{0, 1, ..., \omega\}$ and $t \in \{0, 1, ..., T\}$. The first left and singular right

vectors from the SVD provide the calibration values for b_x and k_t, respectively. In the second step, the obtained sequence k_t is adjusted to ensure the consistency with the correct total number of deaths for each year.

An appropriate time series model for the mortality index k_t is vital to the mortality forecasts through the Lee-Carter model. Lee and Carter (1992) proposed a random walk with drift (RWD):

$$k_{t+1} = k_t + d + e_t, \tag{3}$$

where d is a constant drift and $\{e_t, t \geq 0\}$ are the independent and identically distributed error terms. Both the drift d and the variance of the error terms e_t can be estimated by standard time series estimation techniques. k_t is forecasted stochastically and then used to forecast ASDRs through Equation (1). In addition to the RWD, the Lee-Carter model has been proposed for implementation with the more complex autoregressive integrated moving average (ARIMA) models (e.g., Li, Hardy, and Tan 2009). The ARIMA model can be estimated by the Box-Jenkins method (Box et al., 2015) and implemented using the `auto.arima` function from the `R` package `forecast` (Hyndman and Khandakar 2007).

2.2. Li and Lee's (2005) ACF Model

Li and Lee (2005) pointed out that mortality patterns and trajectories in closely related populations are likely to be similar in some respects. For this reason, they believed that mortality forecasts for individual populations can be improved by taking into account the patterns in a group. To accommodate disparate development in mortality for each individual population, they proposed the ACF model that incorporates the population-specific factors along with a common factor shared by all of the individual populations in the group. Let $m_{x,t,i}$ denote the ASDR of the ith population in the group for age x and year t. The ACF model depicts the mortality development of each individual population in the following form:

$$\log m_{x,t,i} = a_{x,i} + B_x K_t + b_{x,i} k_{t,i} + \epsilon_{x,t,i}, \tag{4}$$

where B_x and K_t constitute the common factor that describes the mortality change for the whole group, $b_{x,i}$ and $k_{t,i}$ are the population-specific components that accommodate disparate development of individual populations, and $\epsilon_{x,t,i}$s are the independent and identically distributed error terms.

As in the case of the Lee-Carter model, constraints need to be imposed so that $\sum_x b_{x,i} = 1$, $\sum_t k_{t,i} = 0$, $\sum_x B_x = 1$, and $\sum_t K_t = 0$ to avoid the unidentifiability issue, and $a_{x,i}$ is calibrated as the average of logarithmic ASDRs over the modeling time period for each population:

$$a_{x,i} = \frac{1}{T+1} \sum_{t=0}^{T} \log m_{x,t,i}. \tag{5}$$

For the calibration of the other parameters, the SVD procedure is applied to the aggregate data of the group to obtain B_x and K_t, and then the SVD procedure is further applied to the residual data, $(\log m_{x,t,i} - a_{x,i} - B_x K_t)$, to get $b_{x,i}$ and $k_{t,i}$ for each individual population.

To obtain the forecast of mortality, Li and Lee (2005) proposed to model K_t with a RWD and $k_{t,i}$ with either a random walk without drift (RW) or a first-order autoregressive (AR(1)) model to guarantee a nondivergent forecast in the long run. After both K_t and $k_{t,i}$ have been modeled, mortality forecasting can be achieved by extrapolating both sequences into years $t > T$ through Equation (4).

Li and Lee (2005) argued that members in a properly chosen group should have similar mortality trends that are sufficiently captured by the common factor so that the remaining population-specific terms would be stable in their scales. They also explained that the model would fail if $k_{t,i}$ has a long-term trend, because the long-term trend might be an indication of a systematically and significantly different trend between population i and the rest in the group.

2.3. Extended ACF Model

The DSA algorithm we develop is a fully data-driven method. To unleash the full potentials of the algorithm, we explore the ways of extending the ACF model to a more general form, and then applied the algorithm to work with the extended version in our numerical studies. Given a group of populations G of size M, the extended ACF model describes logarithmic ASDRs of each individual population $i \in G$ by the following form:

$$\log m_{x,t,i} = a_{x,i} + B_x K_t + \sum_{j=0}^{N_i} b_{x,j,i} k_{t,j,i} + \epsilon_{x,t,i}, \tag{6}$$

where $a_{x,i}$, B_x, K_t, and $\epsilon_{x,t,i}$ carry the same meanings as in the ACF model (4), $(b_{x,j,i}, k_{t,j,i})$ are pairs of population-specific components, and N_i is the number of the population-specific components for population i. Constraints $\sum_x b_{x,j,i} = 1$, $\sum_t k_{t,j,i} = 0$, $\sum_x B_x = 1$, and $\sum_t K_t = 0$ are also imposed to avoid the unidentifiability issue for the extended model.

The strength of the above extended ACF model consists in its flexibility to capture the dynamics of mortality development with multiple population-specific components. The extended ACF model is analogous to the common sparse age–period model of Hatzopoulos and Haberman (2013), which extends the single-population model of Hatzopoulos and Haberman (2009) and introduces multiple orthonormal polynomials of age effects in the linear predictor within a Poisson regression framework for actual number of deaths.

As an integrated part of the above extended ACF model, $a_{x,i}$ is calibrated using Equation (5) as in the ACF mode, but B_x and K_t are calibrated in a different way. In the ACF model, the ASDRs are aggregated across the whole group G using population size as the weighting variable, and B_x and K_t are obtained as the first left and right singular vectors from the SVD procedure applied to the resulting aggregate data (subtracted by the average of log-mortality defined in the same way as Eq. [5]). In contrast, the extended model applies the SVD procedure to $(\log m_{x,t,i} - a_{x,i})$ for each individual population i to obtain $B_{x,i}$ and $K_{t,i}$ as the resulting first left and right singular vectors, and we compute B_x and K_t as the simple average of those $B_{x,i}$'s and $K_{t,i}$ s, respectively; that is,

$$B_x = \frac{1}{M}\sum_{i\in G} B_{x,i} \text{ and } K_t = \frac{1}{M}\sum_{i\in G} K_{t,i}, \tag{7}$$

There followed another round of SVD procedures applied to the residuals, $(\log m_{x,t,i} - a_{x,i} - B_x K_t)$ to calibrate the population-specific components. The first N_i left and right vectors are used to calibrate $b_{x,i,j}$ and $k_{t,i,j}, j = 1, ..., N_i$, respectively.

The above calibration procedure for B_x and K_t is in the same spirit of the model averaging idea in statistics. That is, averaging estimation of the same objective from different sources can potentially reduce the estimation uncertainty. Therefore, we work with the assumption that the individual populations in the group G share the common component pair (B_x, K_t), and the estimation formed by an average of the calibrated value from each individual population can potentially lead to a more efficient estimate of (B_x, K_t).

As an alternative to the simple average in Equation (7), a weighted average using population size can potentially further reduce the estimation uncertainty for the common factor (B_x, K_t), because there is a reason to believe that an estimation from a larger population tends to bring in less uncertainty. However, it is challenging to quantify the precise impact of the population size on the estimation uncertainty of the common factor; in addition, an enhanced performance of the weighted average highly relies upon this piece of information to be precisely incorporated into the calculation. For this reason, we adopt the simple average for the calibration of the common factor in this article.

For mortality forecasting, we fit the sequence K_t with an RWD to forecast the common trend in future mortality changes as in Li and Lee (2005). We allow the population-specific components $k_{t,i,j}$ to embrace the potential benefits of generality by fitting them with the "best" ARIMA model, which is obtained by using the `auto.arima` function from the `R` package `forecast`. We implement the default setting of the `R` function so that a non-seasonal ARIMA model is applied for the data. Parameters *p, d,* and *q* in the model respectively represent the order of the autoregressive regression, the degree of differencing, and the order of the moving average. The fitting procedure proceeds in two stages. The first stage determines d and the second stage determines p and q. A stepwise search is executed at each stage with the initial value being 0 for d and 2 for both p and q. The upper bound is by default set as 2 for d and 5 for both p and q. A detailed description of the ARIMA fitting procedure used by the R function can be found in Hyndman and Khandakar (2007). After obtaining the time series models for these components, we extrapolate them into years $t > T$ to form mortality forecasts for the target population i.

Some clarifications about the above extended ACF model are in order. The parameters M and N_i allow for different levels of model complexity. Tuning M enables the model to have desired size of group, and tuning N_i allows for the desired complexity of population-specific effects. As we will explain later, the rational choice for the parameters is no longer an issue because the values of M and N_i can be learned through a validation step during the modeling procedure. Moreover, this extended model includes the Lee-Carter and the ACF models as special cases. When $N_i = 0$ and $M = 1$ (i.e., group G only contains the target population i), the extended ACF model degenerates into the Lee-Carter model. When $N_i = 1$, the model reduces to the ACF model.

3. DSA-BASED FRAMEWORK FOR MORTALITY FORECASTING

This section introduces our DSA algorithm based framework for mortality forecasting. The framework can work with any multipopulation model, though we will take the extended ACF model for illustration purposes. The DSA algorithm is designed to generate a sequence of groups of populations in increasing size, and a validation procedure is applied to select the optimal group for a given target population. The mortality forecasting is conducted based on the selected optimal group.

In our framework, the data period (years 0 to T) is divided into a modeling period (years 0 to S) and a validation period (years $(S+1)$ to T), and the data set is accordingly divided into a modeling set and a validation set. The DSA algorithm is applied with the modeling set only, and the validation procedure utilizes the validation set to select the optimal group.

3.1. DSA Algorithm: Risk Functions and Moves

The DSA algorithm starts with a group of size one containing the target population only. The algorithm generally utilizes three specific moves or step functions (i.e., deletion, substitution, and addition) to generate a sequence of group selections. For each group size, the algorithm aims to find a group of populations to minimize a prespecified risk function over groups of the same size. Choices of risk function include the mean squared error (MSE), mean absolute error (MAE), mean squared percentage error (MSPE), and mean absolute percentage error (MAPE). Given a group G containing the target population i, we calibrate the extended ACF model (6) with the modeling data set. Denote

$$\log \hat{m}_{x,t,i} = a_{x,i} + B_x K_t + \sum_{j=0}^{N_i} b_{x,j,i} k_{t,j,i},$$

where $N_i \in \{1, 2, ..., N_{\max}\}$ is fixed, and with a slight abuse of notation, all items on the right-hand side mean their calibrated values. Note that $N_{\max}$ denotes the maximum number of population-specific components that we consider in the extended ACF model and we take $N_{\max} = 5$ in our numerical implementation in the sequel.

The risk function on the group G for target population i can be calculated as follows:

- MSE:

$$f(G) = \frac{1}{(\omega+1)\times(S+1)} \sum_{x,t} (\log m_{x,t,i} - \log \hat{m}_{x,t,i})^2 \tag{8}$$

- MAE:

$$f(G) = \frac{1}{(\omega+1)\times(S+1)} \sum_{x,t} |\log m_{x,t,i} - \log \hat{m}_{x,t,i}| \tag{9}$$

- MSPE:

$$f(G) = \frac{1}{(\omega+1)\times(S+1)} \sum_{x,t} \left(\frac{\log m_{x,t,i} - \log \hat{m}_{x,t,i}}{\log m_{x,t,i}} \right)^2 \tag{10}$$

- MAPE:

$$f(G) = \frac{1}{(\omega+1)\times(S+1)} \sum_{x,t} \left| \frac{\log m_{x,t,i} - \log \hat{m}_{x,t,i}}{\log m_{x,t,i}} \right|, \tag{11}$$

where the summation is for x ranging over the considered set of ages and for t over the modeling period (i.e., years 0 to S). The right-hand side of the above formula depends on the group G via B_x and K_t.

With the parameter N_i fixed over the set $\{1, 2, ..., N_{\max}\}$, the proposed algorithm searches to minimize the in-sample value of the adopted risk function (hereafter "risk value" for short) over groups of the same size. For each group size s, the optimal group G_s attains the minimum in-sample risk value among $\mathbb{G}_s$ that denotes the collection of all the subsets of populations in size s from the candidate pool; that is,

$$G_s := \text{argmin}_{G \in \mathbb{G}_s} f(G).$$

It is computationally inefficient to search for the optimal G_s among the whole set $\mathbb{G}_s$ for every s. The DSA algorithm proceeds in a stepwise manner to complete the search. The result, though suboptimal, is more computationally efficient. It follows the same spirit as in many machine learning methods, such as regression tree methods and stepwise (forward or backward) regression procedures, to obtain a good balance between optimality and computational demands.

The DSA algorithm searches through specifically designed iterations and stops with the control of a specific stopping criterion. At each iteration step, the algorithm first maps the current group G of size s into groups of size $s-1$, s, and $s+1$, respectively, through moves or step functions deletion, substitution, and addition. Then, the algorithm solves an optimization problem to secure the optimal outcome throughout all possible results. The three moves are defined as follows:

- **Deletion**: A deletion move allows the removal of one population from the current group. Formally, given the current group G of size s, this move first returns set $\text{DEL}(G)$ that contains s different groups of size $s-1$ by deleting one member from the current group G. Then, the algorithm searches for the optimal element within $\text{DEL}(G)$, denoted as G^- that has the smallest risk value among all possible outcomes; that is,

$$G^- := \text{argmin}_{G' \in \text{DEL}(G)} f(G').$$

- **Substitution**: A substitution move allows the replacement of one population in the current group with another from the remaining set in the candidate pool. Formally, given the current group G of size s, this move first returns set $\text{SUB}(G)$ that contains a number of different groups of size s by making substitutions. The size of $\text{SUB}(G)$ depends on both the value of s and the number of the remaining candidates in the pool. If the number of population in the pool is r in total, then there would be $(r-s)$ remaining candidates and the size of $\text{SUB}(G)$ is $s(r-s)$. In the substation move, the algorithm searches for the optimal element within $\text{SUB}(G)$, denoted as $G^=$, that has the smallest risk value among all possible outcomes; that is,

$$G^= := \text{argmin}_{G' \in \text{SUB}(G')} f(G').$$

- **Addition**: An addition move allows the introduction of one more candidate populations into the current group. Formally, given the current group G of size s, this move first returns set $\text{ADD}(G)$ that contains different groups of size $s+1$ by adding one member into the current group from the rest of the candidates. The size of $\text{ADD}(G)$ equals the number of the remaining candidates in the pool. Then, the algorithm searches for the optimal element within $\text{ADD}(G)$, denoted as G^+, that has the smallest risk value among all possible outcomes; that is,

$$G^+ := \text{argmin}_{G' \in \text{ADD}(G)} f(G').$$

3.2. DSA Algorithm: Ordering of Moves and Pseudo codes

After providing an account of the risk function and the three allowed moves (i.e., deletion, substitution, and addition), we now move on to the ordering of the moves, the initiation, and the stopping criterion of the DSA algorithm. We need the following notations to record information:

- A list G^* to record the best groups of different group sizes. G^*_s, as an element of G^*, represents the best group of size s, $s = 1, 2, \ldots$.
- A vector BEST to record the in-sample risk values corresponding to each element of the list G^*:

$$\text{BEST}(s) = f(G^*_s), \ \ s = 1, 2, ...$$

Throughout the iterations of the algorithm, information contained in vectors G^* and BEST keeps updating untill the algorithm comes to a stop. The details of the algorithm are described as follows:

1. Initialization:
   - The algorithm starts with a group of size one that contains the target population only and retains the rest of the candidate populations in the pool.

- $G_1^* = G_1$, because the best group of size one must be the group containing only the target population.
- The risk value $f(G_1)$ is calculated and BEST(1) is assigned to equal $f(G_1)$.
- A stopping value, called cut-off-growth (COG), is assigned to indicate the maximum number of candidate populations considered for joint mortality modeling. A small COG results in an early stop for the algorithm, whereas a large COG results in a computationally demanding algorithm. COG can be set as large as the size of candidate pool. With consideration having been given to computational feasibility, COG should be set as large as possible because its value corresponds with the class size of candidate groups being considered in the search for the best group.

2. Move through step functions:
   - † Let G be the current working group and denote its size by s.
   - **Deletion** If $s > 3$, search for the optimal updated G^- of size $s-1$ among all allowed deletion moves, where $G^- = \text{argmin}_{G' \in \text{DEL}(G)} f(G')$. If $f(G^-) < \text{BEST}(s-1)$, put $\text{BEST}(s-1) = f(G^-)$, set $G_{s-1}^* = G^-$ to extract the grouping information of G^- to update the optimal group of size $s-1$, update the current working group by $G = G^-$, and return to †.
   - **Substitution** If $s > 2$, find the optimal updated $G^=$ of size s among all allowed substitution moves, where $G^= = \text{argmin}_{G' \in \text{SUB}(G')} f(G')$. If $f(G^=) < \text{BEST}(s)$, put $\text{BEST}(s) = f(G^=)$, set $G_s^* = G^=$ to extract the grouping information of $G^=$ to update the optimal group of size s, update the current working group by $G = G^=$, and return to †.
   - **Addition** Find an optimal updated G^+ of size $s+1$ among all allowed addition moves, where $G^+ = \text{argmin}_{G' \in \text{ADD}(G)} f(G')$. If $f(G^+) < \text{BEST}(s+1)$, put $\text{BEST}(s+1) = f(G^+)$, set $G_{s+1}^* = G^+$ to extract the grouping information of G^+ to update the optimal group of size $s+1$, update the current working group $G = G^+$, and return to †.
3. Stopping criterion: If $s = \text{COG}$, stop the algorithm.

The implementation of the above algorithm will result in a sequence of groups $G^* \equiv \{G_s^*, s = 1, ..., \text{COG}\}$, which gives the optimal choice of grouping for each size $s = 1, ..., \text{COG}$ in the sense to attain the minimum in-sample risk value. Each element in the vector BEST records the in-sample risk value for the corresponding element in the vector G^*. As a concise summary, the DSA algorithm is described by the pseudo-codes in Algorithm 1.

Algorithm 1: DSA algorithm for optimal grouping selection

Input: Target population, candidate pool, COG
Output: G^* (a sequence of groups in increasing size) and BEST (a vector of risks values)
Initiation: $s = 1$, $G = G_1$, $G_1^* = G_1$, $\text{BEST}(1) = f(G_1)$, $\text{BEST}(s) = \infty$ for $s = 2, ..., \text{COG}$;
† Load information: Current group G of size s;
while $s < \text{COG}$ **do**
  if $s > 3$ **then**
    Deletion: Find the optimal G^- of size $s-1$;
    if $f(G^-) < \text{BEST}(s-1)$ **then**
      $\text{BEST}(s-1) = f(G^-)$;
      Set $G_{s-1}^* = G^-, G = G^-$, and return to †;
    end
  end
  if $s > 2$ **then**
    Substitution: Find the optimal $G^=$ of size s;
    if $f(G^=) < \text{BEST}(s)$ **then**
      $\text{BEST}(s) = f(G^=)$;
      Set $G_s^* = G^=, G = G^=$, and return to †;
    end
  end
  Addition: Find the optimal G^+ of size $s+1$;
  if $f(G^+) < \text{BEST}(s+1)$ **then**
    $\text{BEST}(s+1) = f(G^+)$;
    Set $G_{s+1}^* = G^+, G = G^+$, and return to †;
  end
end

3.3. The DSA-Based Prediction Model

For the target population i, the parameters M_i and N_i in the extended ACF model (6) are the group size and the number of population-specific components, respectively. With a fixed $N_i \in \{1, 2, ..., N_{\max}\}$, the implementation of the DSA algorithm yields the "optimal" group of populations, denoted by G^*_{i,M_i,N_i}, for each group size $M_i \in \{1, ..., \text{COG}\}$. For each G^*_{i,M_i,N_i}, we can obtain a mortality model, so implementing the DSA algorithm for $N_i \in \{1, 2, ..., N_{\max}\}$ yields $N_{\max} \times \text{COG}$ mortality models in total.

To determine the best model for prediction, we resort to a validation procedure, which entails the selection of the best values of parameters M_i and N_i. The extended ACF model can be calibrated with the identified G^*_{i,M_i,N_i} from the DSA algorithm for each combination of M_i and N_i. We project the calibrated models into the validation period (i.e., years $S+1$ to T) and calculate the validation risk values using Formulae (8)–(11) with the summation changed to be over the validation period. The optimal M_i and N_i are selected as those that achieve the lowest validation risk value (measured by one of the four metrics calculated as in Eqs. [8]–[11] using validation set). In so doing, we obtain the calibrated model corresponding to the optimal parameters M_i and N_i. Then, the obtained model is projected into the future periods for mortality forecasting.

The DSA-based prediction model can be summarized into a procedure of the following five steps:

1. **Initialization.** Specify the target population i, the corresponding candidate pool (all possible choices of other populations), and COG. Divide the training data of age-specific mortality rates into two periods: a modeling period and a validation period.
2. **Implementation of the DSA algorithm.** For each fixed $N_i \in \{1, 2, ..., N_{\max}\}$, apply the DSA algorithm with the modeling data set to obtain a sequence of optimal groups G_{i,M_i,N_i}, $M_i \in \{1, 2, ..., \text{COG}\}$, that gives the lowest in-sample risk value for each group size M_i.
3. **Validation.** The optimal group G^*_{i,M_i,N_i} emerges by choosing the smallest validation risk among all of the $\text{COG} \times N_{\max}$ models identified from the DSA algorithm.
4. **Parameter Estimation.** With the determined values of M_i and N_i as well as the selected group G^*_{i,M_i,N_i} from the preceding step, we recalibrate the selected model using the whole training data set (including both the modeling and validation data sets).
5. **Forecasting.** We project the recalibrated model into the future periods for mortality forecasting.

4. NUMERICAL ANALYSIS

4.1. Data and Benchmark Models

We apply our DSA-based procedure to a set of populations from the HMD for mortality prediction. The HMD is available for download on the website: https://www.mortality.org/. The HMD contains data from 47 countries or regions, with different scopes and lengths of time. We selected 30 out of the 47; see the Appendix for the detailed list.

We applied the following principles for the choices of populations. The mortality data for 34 countries/regions from 1970 to 2010 were chosen in our study. Our assumption is that the more recent the data, the more relevance they have to current and future mortality. Therefore, our study does not take into consideration the time period before World War II. This proviso may also help to avoid potential impact by rare events or pattern changes. In addition, a moderately sufficient amount of data is preferred to learn a prediction model. That being said, we do our best to maintain a balance between the size of candidate pool and the length of time period for the data. For countries and regions that have mortality data with multiple scopes, only one specific scope is chosen to avoid double counting. Four countries/regions (i.e., Estonia, Iceland, Luxembourg, and Northern Ireland) are left out because their data sets contain many missing data and zero death rates. We use interpolation to fix missing data and replace zero mortality rates with the average value of the same age group from the years preceding and succeeding the given year.

For each of the 30 populations, we consider mortality forecasts for both genders. Each gender of the same population is treated separately; thus, we actually have 60 populations in our study. When one of the 60 populations is the target for prediction, the rest of the 59 populations form the candidate pool, to which we apply the DSA algorithm. We compare our model with the following benchmark models:

- **Lee-Carter**: The Lee-Carter model will be fit for each of the 60 populations independently, using mortality rates in the whole training set. The sequence of k_t obtained from the SVD procedure is fitted using the `auto.arima` function from the `R` package `forecast` to search for a suitable ARIMA model.
- **ACF·AIO**: Each gender consists of 30 populations, and the ACF model is fitted with all 30 populations jointly.

TABLE 1
Summary Statistics of Test Risk Values of the 60 Populations

MSE	First quartile	Median	Mean	Third quartile
Lee-Carter	0.0397	0.0680	0.1409	0.0886
ACF·AIO	0.0291	0.0538	0.0579	0.0794
ACF·GeoInfo	0.0324	0.0629	0.0589	0.0810
ACF·kmeans	0.0318	0.0576	0.0611	0.0773
ACF·kmedian	0.0304	0.0520	0.0583	0.0851
DSA·MSE	0.0233	0.0491	0.0518	0.0751
DSA·MAE	0.0238	0.0494	0.0512	0.0756
DSA·MSPE	0.0276	0.0519	0.0580	0.0811
DSA·MAPE	0.0277	0.0491	0.0548	0.0758
MAE	First quartile	Median	Mean	Third quartile
Lee-Carter	0.1406	0.1644	0.1917	0.1971
ACF·AIO	0.1227	0.1575	0.1567	0.1889
ACF·GeoInfo	0.1237	0.1660	0.1567	0.1838
ACF·kmeans	0.1254	0.1560	0.1554	0.1793
ACF·kmedian	0.1248	0.1558	0.1573	0.1860
DSA·MSE	0.1105	0.1445	0.1465	0.1792
DSA·MAE	0.1104	0.1480	0.1455	0.1751
DSA·MSPE	0.1102	0.1520	0.1558	0.1921
DSA·MAPE	0.1126	0.1444	0.1495	0.1749
MSPE	First quartile	Median	Mean	Third quartile
Lee-Carter	0.0013	0.0026	0.0087	0.0053
ACF·AIO	0.0012	0.0027	0.0048	0.0045
ACF·GeoInfo	0.0011	0.0026	0.0057	0.0052
ACF·kmeans	0.0012	0.0024	0.0054	0.0049
ACF·kmedian	0.0012	0.0026	0.0049	0.0048
DSA·MSE	0.0012	0.0022	0.0055	0.0043
DSA·MAE	0.0012	0.0021	0.0052	0.0047
DSA·MSPE	0.0012	0.0022	0.0056	0.0044
DSA·MAPE	0.0014	0.0023	0.0049	0.0039
MAPE	First quartile	Median	Mean	Third quartile
Lee-Carter	0.0262	0.0353	0.0414	0.0436
ACF·AIO	0.0252	0.0324	0.0343	0.0415
ACF·GeoInfo	0.0247	0.0343	0.0346	0.0433
ACF·kmeans	0.0240	0.0311	0.0347	0.0399
ACF·kmedian	0.0254	0.0312	0.0346	0.0420
DSA·MSE	0.0242	0.0309	0.0330	0.0386
DSA·MAE	0.0235	0.0295	0.0324	0.0394
DSA·MSPE	0.0256	0.0311	0.0344	0.0422
DSA·MAPE	0.0251	0.0312	0.0328	0.0383

- **ACF·GeoInfo**: The 30 populations of each gender are divided into 8 groups based on geographic proximity (for grouping information, see the Appendix). The grouping basically follows Richman and Wüthrich (2018) with a few necessary adjustments, because the populations in our study are not entirely identical to theirs. For each gender, the ACF model is fitted to each geographic group.
- **ACF·kmeans**: For each gender, groups are obtained by a k-means clustering method applied to logarithmic mortality rates data. Then the ACF model is fitted to each group. We set parameter $k = 8$ (the same number of groups as in

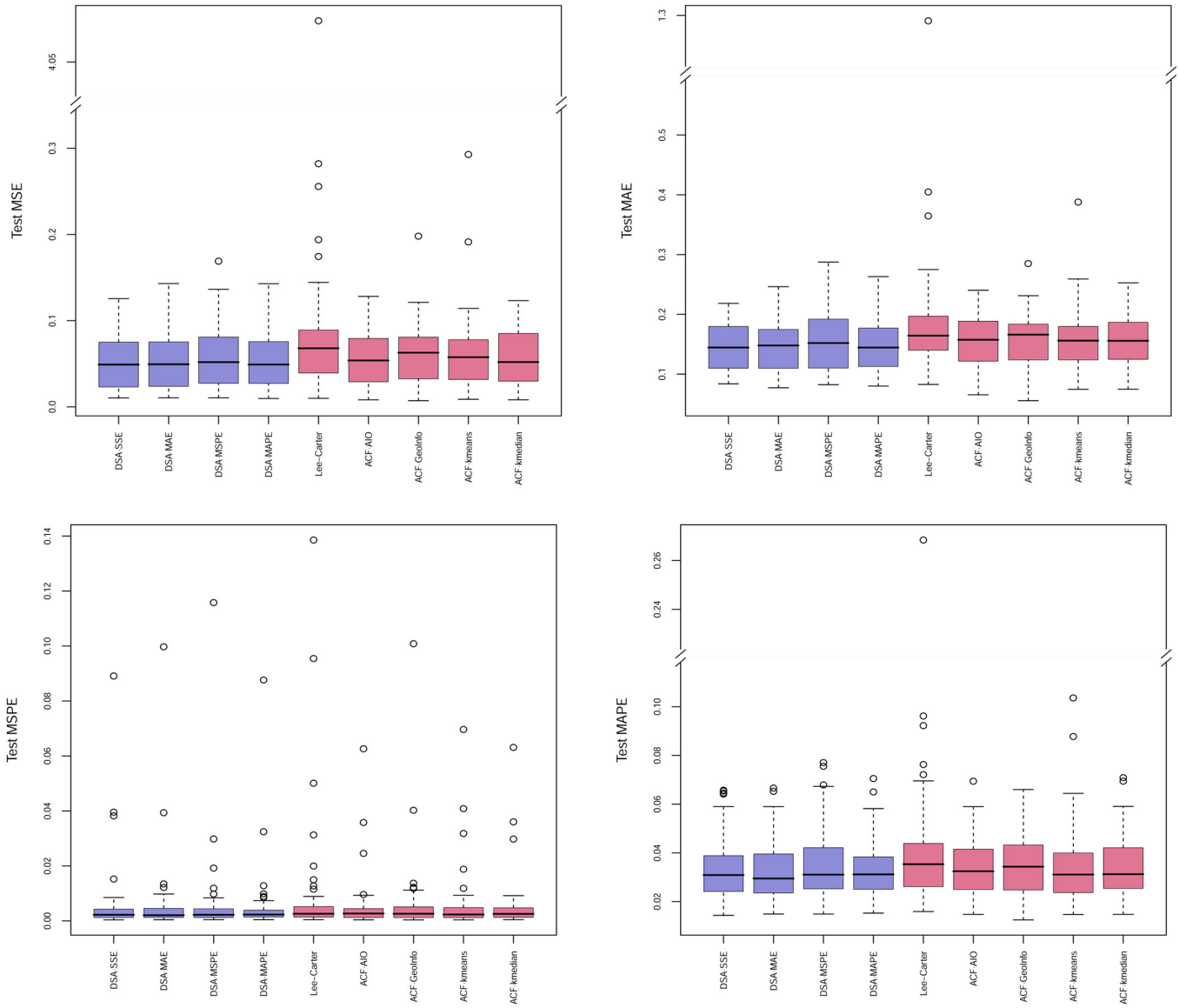


FIGURE 1. Box Plots of Test Risk Values of the 60 Populations.

ACF·GeoInfo) in the clustering algorithm and apply the best clustering result from 1,000 independent random initializations in the clustering algorithm.

- **ACF·kmedian**: The same as ACF·kmeans except that groups are obtained by a k-median clustering method.

Regarding the above benchmark models, two further points are worth mentioning:

(a) Because our DSA-based procedure fits all of the time trend sequences with ARIMA models, we fit an ARIMA model to the sequence k_t obtained from the Lee-Carter model so that the results can be comparable.
(b) The last two benchmark models in the above list are the result of two prevailing clustering algorithms, the k-means and the k-median. It should be noted that k (or K) in these two clustering algorithms is a standard notation in the machine/statistical learning literature, and it differs from the time trend sequence k_t from the Lee-Carter model.

4.2. Numerical Procedure

The numerical procedure is carried out throughout our study in the following steps:

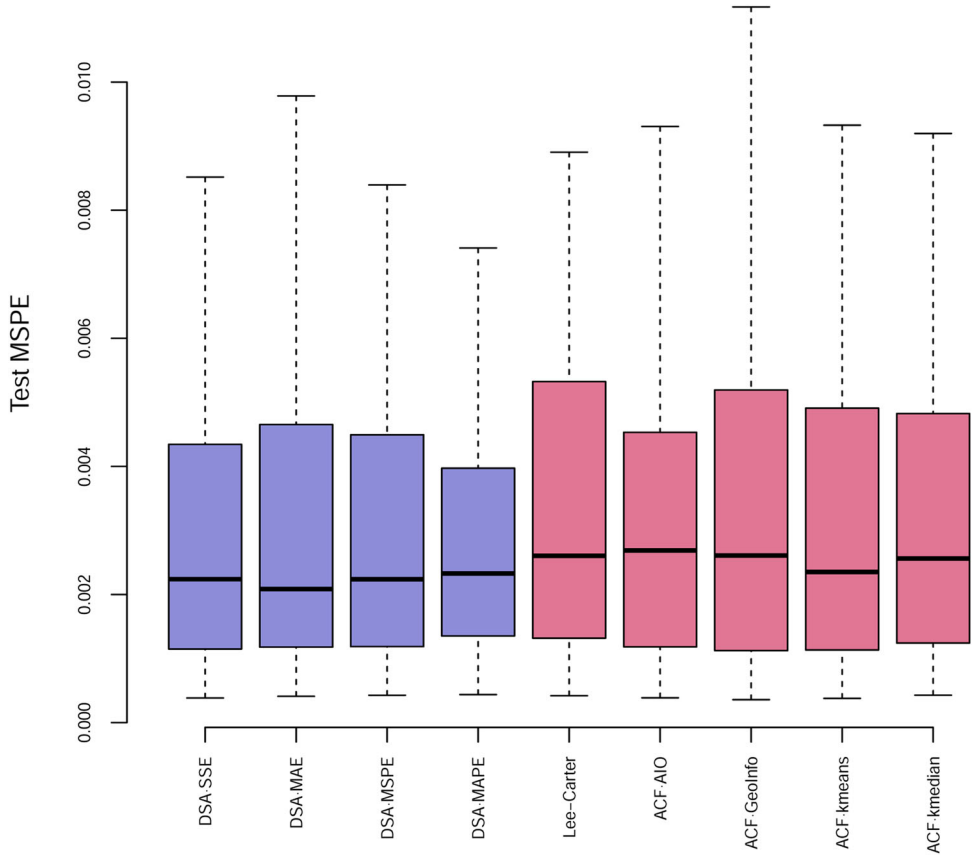


FIGURE 2. Box Plot of Test MSPE Values with Outliers Discarded.

1. We split the mortality data set (consisting of data from 1970 to 2010) into a training set (1970–2002) and a testing set (2003–2010). This gives roughly 80% of the data for training and 20% for testing. The training set is further split into a modeling set (1970–1993) and a validation set (1994–2002).
2. We set each of the 60 populations as the target for prediction and take the remaining 59 gender-specific populations as the candidate pool. The DSA algorithm proceeds with the following specifications:
   (a) We use each of the four metrics defined in Equations (8)–(11) as the risk function in the DSA algorithm. The resulting prediction models are, respectively, labeled as DSA·MSE, DSA·MAE, DSA·MSPE, and DSA·MAPE.
   (b) COG = 30. The COG is set in the DSA algorithm to maintain a balance between model generality and computational demand.
   (c) $N_{\max} = 5$. This parameter controls the complexity of population-specific effects that are contained in the residuals after extracting the common factor in the extended ACF model. The choice of $N_{\max} = 5$ is consistent with the setup of $p = 5$ considered by Booth, Maindonald, and Smith (2002) in the rank-p SVD approximation for mortality modeling.
3. The model validation step is implemented in accordance with the procedure described in Subsection 3.3 to determine the optimal group for each target population. Then, the whole training data set (including both the modeling and validation sets) is used to recalibrate the parameters in the determined model.
4. For each target population, by using the re-calibrated model from the preceding step, we form forecasts of logarithmic ASDRs over the testing period (i.e., 2003–2010). We compute the test risk value for each of the four metrics (i.e., MSE, MAE, MSPE, and MAPE) in Equations (8)–(11) with the summation taken over the testing period.
5. For each benchmark model (i.e., Lee-Carter, ACF·AIO, ACF·GeoInfo, ACF·kmeans, and ACF·kmedian), we calibrate with the training data (including both the modeling and validation sets), extrapolate the calibrated model into the testing period to obtain mortality forecasts, and then calculate risk values for each risk metric as we do with the DSA-based model.

TABLE 2
Wins of One Prediction Model over Another under the Criterion of a p-Value Less than 0.05 from the One-Sided DM Test

MSE	Lee-Carter	ACF·AIO	ACF·GeoInfo	ACF·kmeans	ACF·kmedian
DSA·MSE	(49, 1)	(34, 9)	(30, 12)	(30, 16)	(35, 9)
DSA·MAE	(47, 4)	(37, 10)	(29, 10)	(27, 20)	(36, 11)
DSA·MSPE	(40, 7)	(27, 19)	(19, 20)	(21, 21)	(27, 17)
DSA·MAPE	(42, 5)	(27, 14)	(27, 16)	(25, 21)	(26, 15)
MAE	Lee-Carter	ACF·AIO	ACF·GeoInfo	ACF·kmeans	ACF·kmedian
DSA·MSE	(45, 2)	(34, 13)	(31, 13)	(23, 17)	(36, 10)
DSA·MAE	(46, 5)	(39, 10)	(31,12)	(25, 17)	(39, 10)
DSA·MSPE	(36, 11)	(29, 20)	(23, 21)	(19, 24)	(30, 21)
DSA·MAPE	(43, 5)	(28, 13)	(22, 20)	(21, 17)	(29, 14)
MSPE	Lee-Carter	ACF·AIO	ACF·GeoInfo	ACF·kmeans	ACF·kmedian
DSA·MSE	(29, 9)	(22, 12)	(26, 10)	(23, 13)	(24, 7)
DSA·MAE	(28, 5)	(18, 8)	(24, 9)	(20, 12)	(21, 7)
DSA·MSPE	(26, 7)	(22, 17)	(18, 13)	(16, 18)	(21, 16)
DSA·MAPE	(24, 9)	(22, 12)	(15, 13)	(18, 15)	(22, 13)
MAPE	Lee-Carter	ACF·AIO	ACF·GeoInfo	ACF·kmeans	ACF·kmedian
DSA·MSE	(37, 6)	(34, 15)	(26, 17)	(22, 18)	(34, 13)
DSA·MAE	(36, 4)	(32, 6)	(29, 13)	(23, 16)	(33, 6)
DSA·MSPE	(30, 10)	(25, 17)	(21, 22)	(18, 22)	(25, 17)
DSA·MAPE	(31, 7)	(24, 15)	(20, 19)	(21, 20)	(25, 14)

4.3. Prediction Performance

This Subsection compares the prediction performance of our DSA-based models with that of the benchmarks through summary statistics and a formal hypothesis test. The comparison is based on the test risk values (MSE, MAE, MSPE, or MAPE) from the 60 populations described in Subsection 4.1.

4.3.1. Comparison via Summary Statistics

The numerical procedure in Subsection 4.2 with each prediction model yields a test risk value for each population. A smaller test risk value is more desirable, because it ensures a better performance for a prediction model. Table 1 reports the first quartile, third quartile, mean, and median of the test risk values of the 60 populations obtained under different combinations of risk metric and prediction model. These summary statistics show that our DSA-based prediction models generally yield smaller risk values; that is, they perform better than all benchmarks. In particular, DSA·MSE and DSA·MAE consistently outperform other prediction models in terms of MSE, MAE, and MAPE. Even under the risk metric MSPE, the performance of these two DSA models is comparable to all benchmark models, if not better. The DSA·MSPE model, though having less satisfactory performance than other DSA models, is nevertheless competitive when compared with the benchmark models.

Figure 1 shows box plots of the test risk values of 60 populations under various prediction models evaluated using each risk metric. The test MSPE values are rather dispersive, and a zoomed-in version with all outliers excluded is provided in Figure 2. These figures indicate that the DSA-based models generally yield smaller mean and median, fewer outliers, and smaller variation in the resulting risk values. Therefore, the performance of our DSA model is superior to that of other models, regardless of the risk function that is applied in the DSA step and the metric that is used in calculating the test risk values.

4.3.2. Comparison via Diebold-Mariano Test

For a formal comparison of prediction performance, we conduct the one-sided Diebold-Mariano (DM) test to determine whether the prediction error of a given model is statistically significantly smaller than that of another. The DM test, introduced by Diebold and Mariano (1995) and further improved by Harvey, Leybourne, and Newbold (1997), is a formal statistical hypothesis testing method in the field of forecast comparison. The test is based on two sequences of forecast values from two predictive models. To implement the DM test, we made some modifications to steps 4 and 5 in the numerical procedure

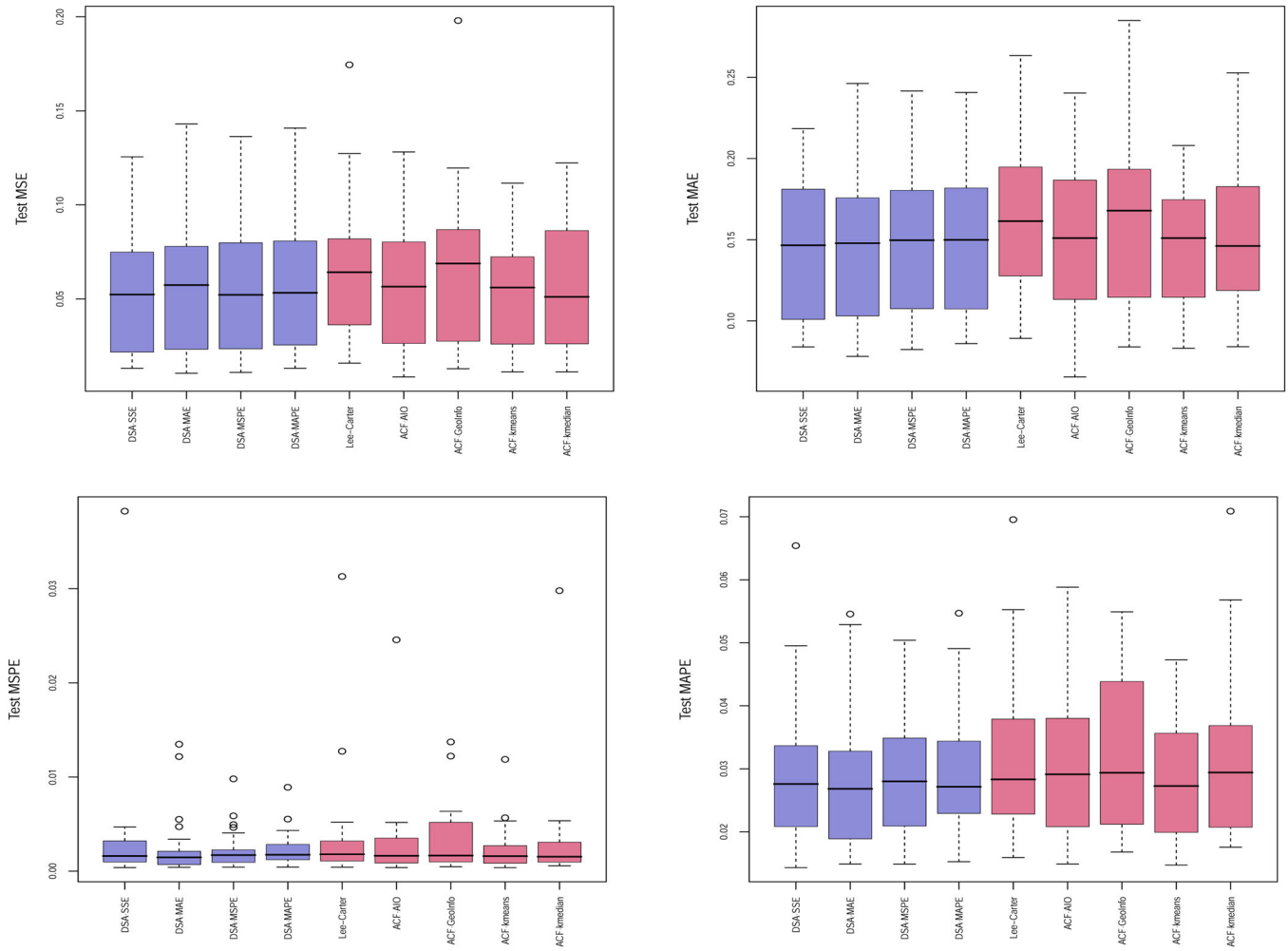


FIGURE 3. Box Plots of Test Risk Values from the 30 Female Populations.

described in Subsection 4.2. We do the summation in Formulae (8)–(11) only for age and calculate risk values for each year over the testing period (2003–2010). This gives us eight test risk values for each prediction model along with each risk metric.

For each metric and each target population, the DM test is conducted pairwise between one DSA model and one benchmark model. To facilitate the exposition of the pairwise test procedure, we label the pair of models in comparison by Model A and Model B. As noted earlier, a sequence of eight test risk values is associated with each prediction model. Two one-sided DM tests are available to us, corresponding to the null hypothesis; that is, Model A is no worse than Model B and vice versa. When Model A is confirmed to be significantly better than the other in the pair by the test with a p value smaller than .05, we count it as a win by Model A over Model B. Similarly, if Model B is confirmed to perform better by the test with a p value smaller than .05, we count it as a win by Model B over Model A. We do the tests for each of the 60 target populations and total all wins obtained by one model over the other. The number of wins by Model A and the number of wins by Model B do not necessarily sum to 60, because the question is still open as to whether Model B necessarily wins when the test does not confirm a win by Model A.

Table 2 reports the number of wins by one model over another under each performance risk metric. Each cell of the table includes two integers. The first integer is the number of wins by the corresponding model in the row over the model on the column, and the second integer is the number of wins by the model in the column over the one in the row. For example, "(49, 1)" in the first cell from the left of the first row in the panel of MSE means that the DSA·MSE model wins 49 times over the Lee-Carter model, and the Lee-Carter model wins over the DSA·MSE model only once among all the 60 comparisons. For the remaining 10 populations, we are unable to deduce which model performs statistically better.

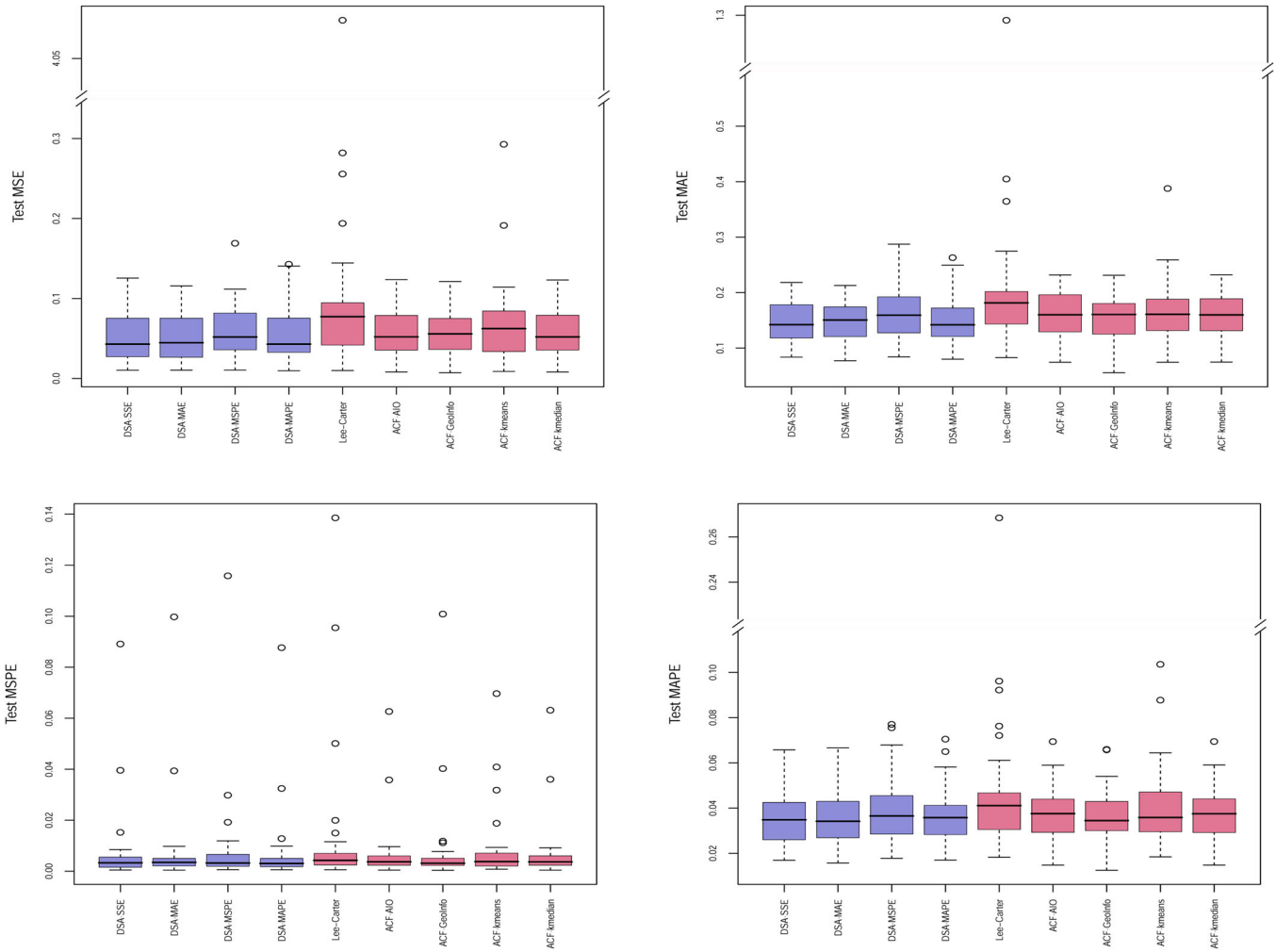


FIGURE 4. Box Plots of Test Risk Values from the 30 Male Populations.

Table 2 clearly indicates, according to the DM test results, the DSA-based models generally perform much better than all of the benchmark models. The superiority of the DSA-based models is all the more evident when compared with the Lee-Carter, ACF·AIO, and ACF·kmedian models, regardless of the applied performance metric. ACF·GeoInfo and ACF·kmeans have the best performance among the set of benchmark models but are not as good as the DSA-based models in general. However, exceptions do occur when the DSA·MSPE model is applied.

4.3.3. Comparison of Gender-Specific Populations

We show the box plots of test risk values for the 30 female populations in Figure 3 and those for the 30 male populations in Figure 4 in order to investigate the performance of the DSA-based model in relation to the benchmark models when applied to gender-specific populations. In general, though the prediction errors for female mortality seem to be stable, those for male mortality rates are volatile, regardless of the risk metric used for performance measure. The prediction quality of every prediction model is higher in the female populations than in the male populations. Nevertheless, the superiority of the DSA-based models over the benchmarks is still evident in the box plots from the populations of either gender.

4.4. Some Further Observations on the DSA-Based Models

Figure 5 demonstrates the distribution of the optimal group size M_i and the population-specific component size N_i selected by each DSA-based model over the 60 populations. The figure shows that the DSA-based models select varying group sizes and component sizes across the 60 target populations. Only a small portion of the 60 populations are assigned with a group

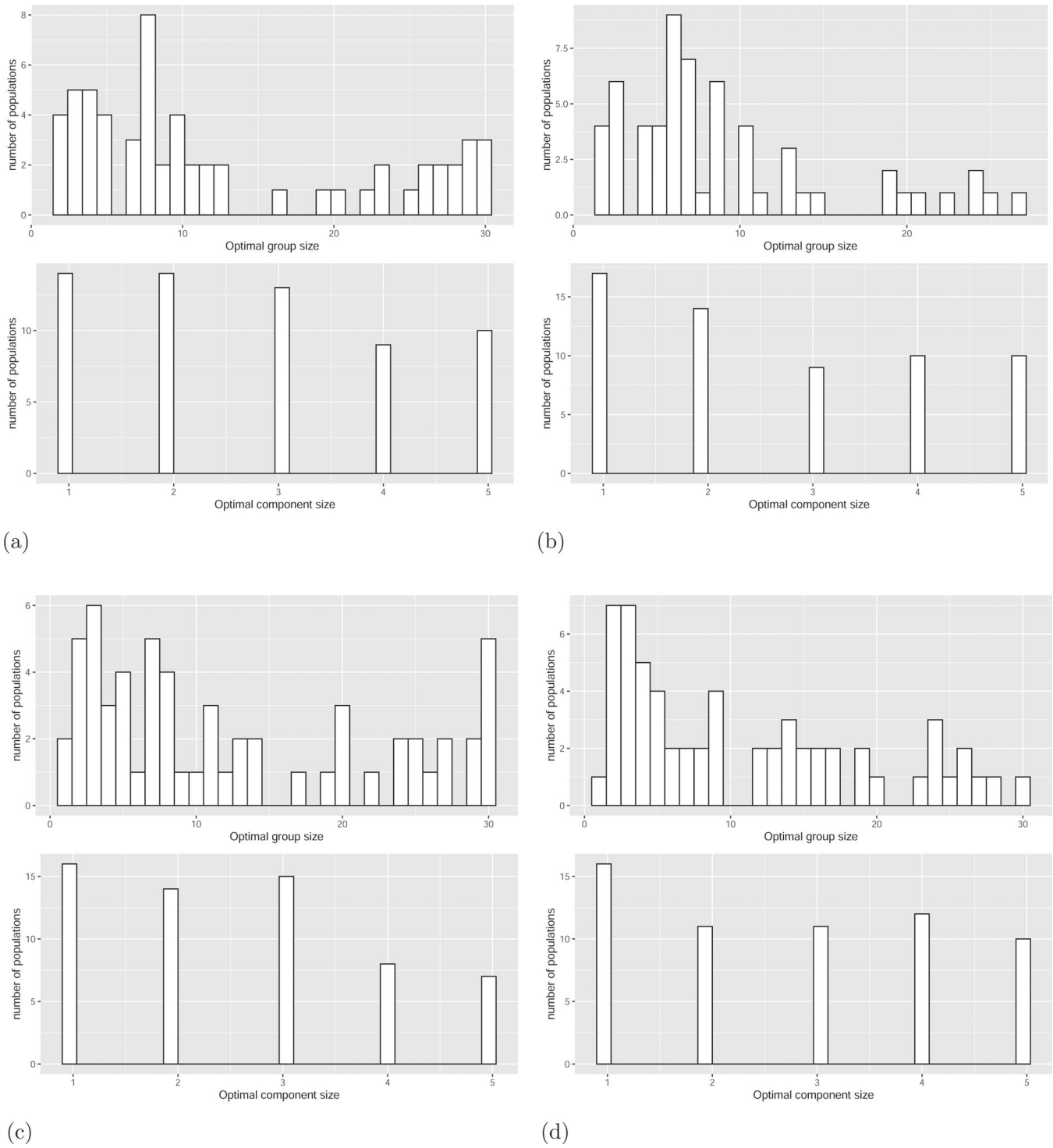


FIGURE 5. Distributions of M_i (Group Size) and N_i (Number of Population Specific Components) Selected by the DSA Based Prediction Model over the 60 Populations: (a) DSA · MSE, (b), DSA · MAE, (c), DSA · MSPE, and (d) DSA · MAPE.

size of one by the DSA-based models. Accordingly, we believe that, in terms of mortality prediction, the DSA algorithm considers the single-population model as less competitive for most of the populations. Moreover, we observe that the DSA-based procedure selects more than one population-specific components (that is, $N_i > 1$) for a majority of the 60 populations. That is,

TABLE 3
Some Examples of Membership in the Optimal Group Selected by the DSA·MSE Model

Target population	Members	Test MSE	
		DSA·MSE	ACF·GeoInf
Bulgaria.female	Bulgaria.female	0.0481	0.0961
	Australia.female		
	Czech Republic.female		
	Denmark.male		
	Japan.female		
	Latvia.female		
	Latvia.male		
	Norway.female		
	Russia.male		
	Slovakia.male		
	New Zealand.male		
	Scotland.male		
Hungary.female	Hungary.female	0.0529	0.0853
	Czech Republic.male		
Canada.female	Canada.female	0.0168	0.0198
	Canada.male		
	Austria.male		
	Japan.male		
	Japan.female		
	Netherlands.male		
	Taiwan.female		
	U.S.A.female		
Taiwan.male	Taiwan.male	0.0199	0.0360
	Switzerland.male		
	Portugal.female		
	Czech Republic.male		
	Japan.female		

the extension that includes more than one population-specific component in Model (6) helps to enhance the performance of mortality prediction.

We also explore the membership of the optimal group selected by the DSA-based models for each target population, and conduct a comparison with the grouping information from the geographic proximity specified in the Appendix. For the sake of brevity, we focus on the DSA·MSE model and present four specific examples in Table 3. The membership results ensure that our DSA-based procedure not only produces groups that are consistent with geographic information to certain degree but also makes further improvement. In some cases, when the group based on geographic proximity is satisfactory, our DSA-based procedure proceeds to choose a further subset as the optimal one. For instance, the group of female Hungarians belongs to this category. In other cases, discrepancies occur between the optimal group from the DSA procedure, and the geographic proximity information, then several representative populations from other parts of the world are added to the optimal group. For example, female Taiwanese are added to the group of female Canadians. This happens when the prediction accuracy can be enhanced with the group that contains not only geographically proximate populations but also distant ones with similar mortality development patterns. Furthermore, some target populations have relatively unstable mortality patterns. In order to offset the adverse effects of volatility in the mortality rates of the target population and avoid obtaining unstable prediction results, some populations from other parts of the world that are not geographically proximate are added to the optimal group. One such example is the Bulgarian female population.

5. DISCUSSION AND CONCLUDING REMARKS

It has been well recognized that an enhanced mortality prediction accuracy can be achieved through joint mortality modeling of human mortality data. It is precisely by virtue of joint modeling that our data-driven framework selects populations from any given candidate pool to enhance mortality forecasting of individual populations. The DSA algorithm is the key element of our framework, designed to screen populations for joint modeling. The framework is fully data driven and flexible enough to embrace any multipopulation mortality model. Numerical analysis with the HMD is conducted to confirm that the performance of the proposed DSA-based prediction method is indeed superior to that of many prevailing benchmark models.

The implementation of the DSA algorithm in our framework yields a sequence of groups of increasing size. Thus, the algorithm can be treated as a stepwise forward procedure for the selection of populations. It starts with a group consisting exclusively of the target population. Then, more populations are added to gradually increase the group size, though deletion and substitution moves might be repeated in the course of the procedure. An alternative to the DSA algorithm is the addition-substitution-deletion (ASD) algorithm, which reverses the order of the three moves in the DSA algorithm. Different from the DSA algorithm, the ASD algorithm starts with a group composed of the target population and all of the candidates from the pool. Then, the group size is gradually decreased by removing certain populations from the group. Both the DSA and ASD algorithms can eventually provide a sequence of grouping results for model selection in the validation step. Instead of the ASD algorithm, we choose the DSA algorithm for our mortality forecasting framework largely because of its computational efficiency. The DSA algorithm allows the users to cut off the search process by setting the parameter COG in Algorithm 1 and control the computational demand in running the algorithm. When a COG smaller than the total size of the input candidate pool is set, the algorithm will search for optimal groups in size only up to the designated value of COG and discard all groups larger than COG. On the contrary, if we need to control the computational demand in the ASD algorithm and set a value for COG smaller than the total number of the candidate pool, the algorithm will leave out all groups smaller than COG in the search for the best grouping.

Quantification of uncertainty is one of the essential aspects of any forecasting procedure. With respect to the forecasting of mortality rates, fan charts are commonly employed for uncertainty quantification. Fan charts delineate confidence intervals for future mortality rates over time. Examples can be found in Hatzopoulos and Haberman (2009), Li, Hardy, and Tan (2009) and Cairns et al. (2011). The implementation of our DSA framework yields a joint model of mortality rates for the target population along with a set of selected auxiliary populations. We can simulate a great number of paths for future mortality rates from the resulting mortality model, and then generate fan charts from the simulated paths. Because it is quite a conventional procedure to build fan charts and the most innovative part of our DSA-based framework lies in its flexibility to embrace various joint models and information from a given candidate pool of populations, we choose not to delve into the details of quantification of forecast uncertainty in this article.

When conducting a comprehensive evaluation of a mortality forecasting model, many factors are worthy of our attention, and the empirical prediction accuracy is merely one of them. For instance, Cairns. (2009) and Haberman and Renshaw (2011), among others, have considered various desirable properties, such as parsimony, transparency, generation of simple paths, cohort effects, nontrivial correlation structure, robust parameter estimates, biological reasonableness. It is only in terms of the empirical prediction accuracy that this article asserts superiority of the DSA-based models over the benchmarks. On no account do we claim the superiority of our model in any other aspect. For example, the ACF model using stationary times series models, including the random walk with zero drift and AR(1) model, for the population-specific components, can yield coherent mortality forecasting results. By extending the models to the general ARIMA model, the resulting model can no longer generate coherence.

Many interesting aspects of the DSA-based mortality forecasting framework are worth further exploration. Because the DSA algorithm accommodates itself to any joint mortality models, it would be interesting to investigate the performance of the DSA-based model along with other available multiple-population models. For example, we may apply the DSA algorithm with the Cairns-Blake-Dowd model proposed by Cairns, Blake, and Dowd (2006) or the common sparse age–period mode introduced by Hatzopoulos and Haberman (2013). Likewise, an exploration of our DSA-based procedure for the prediction of mortality rates of a particular age group (i.e., infant, teenage, or older age groups) can also be rewarding. Moreover, a challenging yet promising venture is to extend the DSA-based procedure to select not only population groups but also time periods and age bands. In so doing, information can be borrowed to make prediction of mortality of a target age (or a small set of ages) for a target population. In addition, machine learning methods have been introduced into the area of mortality forecasting. For example, Richman and Wüthrich (2018) extended the Lee-Carter model to multiple populations by using neural networks for automatic selection of optimal model structure, and Nigri et al. (2019) applied a recurrent neural network with a long short-term memory architecture to the Lee-Carter model for improved predictive capacity. In short, a comparison or combination of the DSA-based procedure and other machine learning techniques can be potentially effective in mortality forecasting.

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APPENDIX A. GEOGRAPHIC GROUPING

The following table illustrates the grouping of populations using geographic information.

Table A.1 classifies the 30 populations into eight geographic groups:

TABLE A.1
Geographic Grouping Information

Target country	Geographic group	Target country	Geographic group
Australia	Oceania	The Netherlands	Western Europe
Austria	Western Europe	New Zealand	Oceania
Belarus	Eastern Europe	Norway	Scandinavia
Belgium	Western Europe	Poland	Eastern Europe
Bulgaria	Eastern Europe	Portugal	Southern Europe
Canada	North America	Russia	Eastern Europe
Czech Republic	Eastern Europe	Slovakia	Eastern Europe
Denmark	Scandinavia	Spain	Southern Europe
Finland	Scandinavia	Sweden	Scandinavia
France	Western Europe	Switzerland	Western Europe
Hungary	Eastern Europe	Taiwan	Asia
Italy	Southern Europe	England and Wales	Great Britain
Japan	Asia	Scotland	Great Britain
Latvia	Eastern Europe	United States	North America
Lithuania	Eastern Europe	Ukraine	Eastern Europe

- Oceania (two members): Australia, New Zealand
- North America (two members): Canada, United States
- Great Britain (two members): England and Wales, Scotland
- Asia (two members): Japan, Taiwan
- Scandinavia (four members): Denmark, Finland, Norway, Sweden
- Western Europe (five members): Austria, Belgium, France, Switzerland, The Netherlands

- Eastern Europe (10 members): Belarus, Poland, Bulgaria, Russia, Czech Republic, Slovakia, Hungary, Latvia, Lithuania, Ukraine
- Southern Europe (three members): Spain, Italy, Portugal.